1. Item 1 (Most Relevant)

Perform a simple calculation

📌 **Why this is central:**
Solving this problem relies almost entirely on a **direct arithmetic calculation**:

- Convert 9 km to centimetres
- Divide the total distance by the number of steps

📌 **Core calculation:**

$$\text{Average step length} = \frac{900{,}000 \text{ cm}}{22{,}500}$$

➡️ No complex modelling or reasoning is required; this is a straightforward "total ÷ unit count" task.

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2. Item 12

Use a multi-step procedure leading to a mathematical solution

📌 **Although simple, it involves several steps:**

1. Recognise that an average step length is required
2. Convert distance units from kilometres to centimetres
3. Divide total distance by total steps
4. Obtain a numerical result

📌 **This forms a clear, linear solution process**

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3. Item 13

Use an understanding of the context to guide or expedite the mathematical solving process

📌 **Role of contextual understanding:**

- The question asks for an **estimate**, not an exact value
- Excessive precision is unnecessary
- Knowing that a typical step length is on the order of several tens of centimetres helps check the reasonableness of the result

📌 **For example:**

- Results such as 4 cm or 400 cm should immediately appear unreasonable

4. Item 5

Understand and utilise constructs based on definitions, rules and formal systems, including employing familiar algorithms

📌 **Relevant concepts and rules:**

- Definition of an average
- Unit conversions (1 km = 1000 m; 1 m = 100 cm)
- Standard division algorithms

📌 **All are familiar mathematical tools**

5. Item 7

Manipulate numbers, graphical and statistical data and information

📌 **Numerical manipulation includes:**

- Working with large numbers (e.g. 900,000)
- Performing division
- Simplifying numerical expressions

📌 **No statistical inference is involved, but flexible number handling is required**

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6. Item 8

Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

📌 **In a written solution, students would typically:**

- Show the unit conversion
- Present the final answer clearly with correct units (cm)

📌 **The emphasis is on communicating the result rather than justifying a model**

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7. Item 11

Switch between different representations in the process of finding solutions

📌 **Representation changes include:**

- Converting distance representations (km → cm)
- Moving from a total quantity to a unit quantity (per step)

📌 **These changes are simple but genuine representation shifts**

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8. Item 10

Make sense of, relate and use a variety of representations when interacting with a problem

📌 **Representations used:**

- Numerical values (9 km, 22,500 steps)
- Measurement units (km, cm, steps)

📌 **However, the variety of representations is limited**

9. Item 3

Implement a given strategy to determine a mathematical solution

📌 **The strategy is essentially given in the question:**

- "Average step length" clearly indicates the use of an averaging procedure
- Students mainly execute the strategy rather than choose it

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10. Item 2

Select an appropriate strategy from a list

📌 **Lower relevance because:**

- The solution path is obvious
- There is no meaningful comparison between alternative strategies

---

11. Item 9

Use mathematical tools, including technology, simulations and computational thinking

📌 **Reason for low relevance:**

- No calculator, digital tools, or simulations are required
- The problem can be solved by hand

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12. Items 4 and 6

Make or develop mathematical diagrams, graphs, or constructions

📌 **Reason:**

- No diagrams or graphs are needed
- No information needs to be extracted from visual representations

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13. Item 14 (Least Relevant)

Make generalisations based on the results of applying mathematical procedures

📌 **Reason:**

- The task requires a single numerical estimate
- No generalisation or extension is involved

① 第 1 条（最核心）

Perform a simple calculation

📌 **核心原因：**
这道题的求解几乎完全依赖于一个**直接的算术计算**：

- 将 9 km 转换为 cm
- 用总路程除以总步数

📌 **计算本质：**

$$\text{平均步长} = \frac{900\,000\ \text{cm}}{22\,500}$$

➡️ 不涉及复杂推理或建模，是标准的“已知量 → 求单位量”计算

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② 第 12 条

Use a multi-step procedure leading to a mathematical solution

📌 **虽然简单，但仍是多步骤：**

1. 识别所需的是“平均步长”
2. 将距离单位从 km 转为 cm
3. 用总距离 ÷ 步数
4. 得到数值结果

📌 **这是一个清晰、线性的解题流程**

---

③ 第 13 条

Use an understanding of the context to guide or expedite the mathematical solving process

📌 **情境理解的作用：**

- 题目要求的是 **estimate（估计）**
- 结果只需合理，不必追求过高精度
- 知道“步长”通常在几十厘米范围内，有助于检验答案是否合理

📌 **例如：**

- 如果算出 4 cm 或 400 cm，学生应意识到不合理

④ 第 5 条

Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

📌 **涉及的基本概念与规则：**

- 平均值的定义
- 长度单位换算（1 km = 1000 m，1 m = 100 cm）
- 除法算法

📌 **都是学生熟悉的数学工具**

⑤ 第 7 条

Manipulate numbers, graphical and statistical data and information

📌 **数值操作包括：**

- 大数（900 000）的处理
- 整数除法
- 数值简化

📌 **不涉及统计推断，但涉及数的灵活处理**

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⑥ 第 8 条

Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

📌 **在书面解答中通常需要：**

- 展示单位转换
- 给出最终结果并标明单位（cm）

📌 **强调表达“答案是什么”而不是“为什么这样建模”**

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⑦ 第 11 条

Switch between different representations in the process of finding solutions

📌 **表示转换体现在：**

- 距离的不同表示形式（km → cm）
- 从“总量”到“单位量（每一步）”

📌 **虽然转换不复杂，但确实存在**

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⑧ 第 10 条

Make sense of, relate and use a variety of representations

📌 **使用的表示包括：**

- 数值表示（9 km，22 500 steps）
- 单位表示（km、cm、步）

📌 **但表示形式较少，复杂度不高**

⑨ 第 3 条

Implement a given strategy to determine a mathematical solution

📌 **策略已隐含在题目中：**

- “average step length” 明确暗示用平均值计算
- 学生主要是执行，而非选择策略

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⑩ 第 2 条

Select an appropriate strategy from a list

📌 **相关性较低原因：**

- 解题路径非常明确
- 不存在多种可行策略的比较或选择

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⑪ 第 9 条

Use mathematical tools, including technology, simulations and computational thinking

📌 **原因：**

- 不需要计算器、技术工具或模拟
- 手算即可完成

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⑫ 第 4 / 6 条（相关性极低）

Make or develop mathematical diagrams, graphs, constructions

📌 **原因：**

- 完全不需要图形或图表
- 也无需从图中提取信息

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⑬ 第 14 条（最低）

Make generalisations based on the results of applying mathematical procedures

📌 **原因：**

- 题目只要求一个具体数值
- 不涉及推广或规律总结